

Mina Avazbeigi

Research Assistant

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Professional Summary

AI researcher with hands-on experience in **computer vision**, **medical image processing**, and **NLP**. Developed neural architectures for medical applications and industrial predictive maintenance. With publications in medical image processing, NLP and optimization. Committed to bridging the gap between academic research and real-world applications.

Education

MSc in Computer Engineering - Shahid Bahonar University of Kerman, Iran, Kerman

Sep 2019 - Sep 2022

- **Thesis:** Created a novel MR image preprocessing pipeline, for semantic temporal brain tumor classification. **Focus areas:** Machine learning, Deep learning, Computer vision. **GPA:** 17.82 (out of 20)
- **Advisor:** Dr. Ghazizadeh-Ahsaei.

BSc in Computer Engineering - Shahid Bahonar University of Kerman, Iran, Kerman

Dec 2011 - Jan 2017

- **Thesis:** Implemented a Comprehensive B2B Marketing Platform for Industrial Enterprises using MVC and SOA Architecture". **Focus areas:** Software development. **GPA:** 14.13 (out of 20)
- **Advisor:** Dr. Ghazvini.

Academic Experience

Research Assistant in Machine Learning – (part-time) Iran University of Medical Sciences, Iran, Tehran

Oct 2025 – Present

Topic Modeling: Developing machine learning models for topic modeling in Regenerative Medicine technology analysis. Published in: Journal of Medical Library and Information Science

Research Assistant in Machine Learning – (part-time) Kerman University of Medical Sciences

Apr 2023 – Aug 2025

- **Data collection and Text Classification:** Developed CNN-based fact checking methods for COVID-19 text classification. Published in: BMC Medical Informatics and Decision Making
- **Topic modeling in oncological studies:** Developed and optimized topic modeling methods for breast cancer scientific publications.

Lecturer - (part-time) Technical and Vocational University of Kerman, Iran, Kerman

Sep 2023 - Jul 2024

- Designed and taught courses in Data Science and Web Programming.
- Supervised several students.

Master Thesis Research - Shahid Bahonar University of Kerman, Iran, Kerman

Mar 2020 - Sep 2022

- **Computer vision:** Developed 3T MRI preprocessing pipeline for gliomas tumor classification, eliminating segmentation process while improving diagnostic accuracy. Adapted RNNs to sequential images. Published in: SN Computer Science

Teacher Assistant - Shahid Bahonar University of Kerman, Iran, Kerman

Assisted in B.Sc. Computer Science courses across three semesters:

- Dr. Satari (one semester) – Lectured & graded coursework for 30 students.
- Dr. Ghazvini (two semester) – Led lectures & grading for 40 students.

Oct 2019 - Dec 2019
Sep 2013 - Apr 2014

Publications

- "Lion Optimization Algorithms (LOAs): A Review on Theory, Variants, and Applications" IJISCE, 2023. [\[Link\]](#) (first author)
- "A Complete Preprocessing Pipeline for Brain Tumor Classification and Overall Survival Prediction Using Bidirectional Convolutional LSTM" SN Computer Science, 2025. [\[DOI\]](#) (first author)
- "Classifying and Fact-Checking Health-Related Information About COVID-19 on Twitter Using TextConvoNet" BMC Medical Informatics and Decision Making, 2025. [\[DOI\]](#)
- Global Innovation Dynamics in Regenerative Medicine: A Patent-Based Assessment. Journal of Medical Library and Information Science, 2026. [\[DOI\]](#)

Skills

- **Machine Learning & AI:** Computer Vision, NLP, Predictive Modeling, Time-Series Analysis, Image processing
- **Tools & Frameworks:** Python & C#, Git, Colab
- **Languages:** Persian (Fluent), **English (C1: IELTS overall score 7)**

Additional & Transferable Experiences

Machine Learning Engineer – (part-time) Kerman Railways, Iran, Kerman

Mar 2024 - Nov 2024

- **Time-series analysis:**
Designed AI-driven predictive maintenance models for railways infrastructure using RNN-based time-series analysis. Designed first AI-Driven Decision Support System for real-time identification of railway degradation and maintenance prediction in Iran.

Volunteer Reviewer - Applied Intelligence Journal

Feb 2023 – Mar 2024

- Reviewed research papers in AI and machine learning applications.

Python Developer - Shahid Bahonar University of Kerman, Iran, Kerman

Jan 2022 - Nov 2022

Projects (Related & Transferable)

- **Topic modeling for Technology analysis** in Regenerative Medicine Using BERTopic
- **Railway Maintenance Predictor** – Built RNN model for time-series based anomaly detection for infrastructure monitoring automation.
- **BERT-based Humor Detection in Medical Texts** – Fine-tuned BERT on short, ambiguous COVID tweets
- **CNN-based COVID-19 fact-checking and misinformation detection**
- **Trustworthy Dataset** – Co-developed COVID-19 misinformation detection dataset [publicly available at [GitHub](#)]
- **COVIDFactChecker Dataset** – Co-developed Twitter misinformation detection dataset [publicly available at [GitHub](#)]
- **LDA Topic modeling** – Designed Topic modeling methods on oncological studies
- **Persian topic modeling** – Developed ML methods on Persian contents.

Awards & Scholarships & Honors

- Awarded two years MSc government Scholarship at Shahid Bahonar University of Kerman (2019–2021)

References

Dr. Homa Arshadi - Assistant Professor of Medical Librarianship and Information Sciences. School of Health Management & Information Sciences, Iran University of Medical Sciences, Iran, Tehran. Email: Arshadi.h@iums.ac.ir

Dr. Mostafa Ghazizadeh-Ahsaee - Assistant Professor of Computer Engineering
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